

Hybrid transformer-based model for mammogram classification by integrating prior and current images

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Abstract

Background: Breast cancer screening via mammography plays a crucial role in early detection, significantly impacting women's health outcomes worldwide. However, the manual analysis of mammographic images is time-consuming and requires specialized expertise, presenting substantial challenges in medical practice.

Purpose: To address these challenges, we introduce a CNN-Transformer based model tailored for breast cancer classification through mammographic analysis. This model leverages both prior and current images to monitor temporal changes, aiming to enhance the efficiency and accuracy (ACC) of computer-aided diagnosis systems by mimicking the detailed examination process of radiologists.

Methods: In this study, our proposed model incorporates a novel integration of a position-wise feedforward network and multi-head self-attention, enabling it to detect abnormal or cancerous changes in mammograms over time. Additionally, the model employs positional encoding and channel attention methods to accurately highlight critical spatial features, thus precisely differentiating between normal and cancerous tissues. Our methodology utilizes focal loss (FL) to precisely address challenging instances that are difficult to classify, reducing false negatives and false positives to improve diagnostic ACC.

Results: We compared our model with eight baseline models; specifically, we utilized only current images for the single model ResNet50 while employing both prior and current images for the remaining models in terms of accuracy (ACC), sensitivity (SEN), precision (PRE), specificity (SPE), F1 score, and area under the curve (AUC). The results demonstrate that the proposed model outperforms the baseline models, achieving an ACC of 90.80%, SEN of 90.80%, PRE of 90.80%, SPE of 90.88%, an F1 score of 90.95%, and an AUC of 92.58%. The codes and related information are available at <https://github.com/NabaviLab/PCTM>.

Conclusions: Our proposed CNN-Transformer model integrates both prior and current images, removes long-range dependencies, and enhances its capability for nuanced classification. The application of FL reduces false positive rate (FPR) and false negative rates (FNR), improving both SEN and SPE. Furthermore, the model achieves the lowest false discovery rate and FNR across various abnormalities, including masses, calcification, and architectural distortions (ADs). These low error rates highlight the model's reliability and underscore its potential to improve early breast cancer detection in clinical practice.

KEYWORDS

CNN, prior and current mammograms, transformer

1 | INTRODUCTION

Breast cancer is the second leading cause of cancer-related deaths among women in the United States, constituting 30% of all female cancer diagnoses.¹ The American Cancer Society reported approximately 276 480 new cases and 43 600 deaths from breast cancer in 2021. These figures are estimates that take into account challenges, such as reporting delays, incomplete data, diagnostic variability, population mobility, and sampling errors. The incidence of breast cancer has been increasing by approximately 0.5% each year.² This trend is mirrored in China, where breast cancer rates are rising, and the age of onset is decreasing each year.³ Globally, breast cancer is the most diagnosed non-cutaneous cancer in women, with over 2 million cases each year,⁴ highlighting the urgency for early detection strategies to curb the mortality rate associated with breast cancer.

Scientists have developed screening mammography as a vital technique for early breast cancer identification, with the goal of substantially decreasing the mortality rate. Studies show that mammography screening may result in around a 40% decrease in death rates from breast cancer.⁵ Although advantageous, the procedure has significant constraints. For example, 15 out of every 1000 women who get mammography screening will be advised to have a needle biopsy because of questionable results.⁶ About 10 to 13 of these biopsies will show no malignancy, resulting in many false positives. Kuhl et al.⁷ have highlighted the underdiagnosis of important breast malignancies as a serious drawback of mammography screening. Various methods have been implemented to improve the effectiveness of mammography screening. The procedures include double-reading mammograms, doing annual screenings⁸, obtaining two distinct images (the Cranio-Caudal [CC] and the mediolateral oblique [MLO]) of each breast⁹, and comparing current mammography with earlier ones.¹⁰ These approaches aim to enhance the identification of crucial signs of breast cancer, such as masses, microcalcifications (MCs), ADs, and asymmetries, which act as biomarkers for the illness or its risk. However, manual detection of these features is costly and burdens the small workforce of breast imaging radiologists.¹¹ Therefore, a computer-aided diagnosis (CAD) system is essential for assisting clinicians in making diagnoses, which may save time and decrease the possibility of incorrect diagnoses¹².

Mammogram classification is vital in CAD systems for interpreting mammography images and has historically relied on conventional machine learning

techniques^{13–17}. However, these techniques largely depend on manual feature selection and engineering, which can introduce subjectivity and require extensive domain expertise, making the process time-consuming and potentially overlooking critical information. In the past decade, convolutional neural networks (CNNs), a type of deep learning, have significantly improved CAD systems^{18–23} by enabling the automatic extraction of features from mammogram images. These approaches^{18–20,23} have mostly been utilized on full-field digital mammography (FFDM) images, often concentrating on a single view or using smaller image segments to mitigate computational load. Yet, the precision (PRE) in detecting and classifying breast abnormalities through these advanced imaging techniques continues to face significant hurdles. The complex representation of these abnormalities, which closely imitate the nearby tissues, poses a significant difficulty in differentiating small or poorly defined tumors. This difficulty is made harder by the intrinsic properties of mammograms, which are high-resolution images rich in detail, demanding the use of complex computational methods for their thorough analysis.

Several CNN-based studies^{24–26} have adopted multi-view approaches,²⁷ specifically using both CC and MLO views, to tackle the challenge of detecting and classifying subtle breast abnormalities accurately. Carneiro et al.²⁴ utilized binary segmentation masks to enhance accuracy (ACC), while Dhungel et al.²⁵ and Khan et al.²⁶ expanded this to two and four views, respectively. However, these methods rely on regions of interest (ROIs) or segmentation masks that require annotation by highly trained physicians. In contrast, other methods^{20,27–30} avoid the need for pixel-level annotations. For instance, Wu et al.²⁰ used heatmaps, and Song et al.,³¹ Zhang et al.,²⁸ and Quy et al.²⁹ refined classification methods with reduced annotation requirements. Despite these advances, CNN-based approaches still struggle with capturing long-distance dependencies due to their limited receptive fields, which hampers their ability to integrate global information effectively. Xia et al.³⁰ tackled this issue by integrating ResNeXt with 12 Transformer blocks, enhancing the feature integration of both local and global features. However, this solution increased model complexity and computational demands.

Comparing current and prior mammograms is crucial for improving breast cancer detection and classification,³² as it allows radiologists to identify subtle changes over time. Several deep learning models trained on both current and historical mammograms have shown superior performance.^{27,33,34} For instance, Kooi and Karssemeijer³³ used paired CNNs to examine

current and prior mammograms, although this approach relies heavily on radiologists to identify and map ROIs. Park et al.²⁷ developed a multiview classification model that improves diagnosis ACC using both current and prior mammograms but requires significant computational resources and rigorous synchronization across multiple views. Bai et al.³⁴ used a Siamese-based network to compare features between current and prior mammograms but highlighted the limitations of CNNs in recognizing long-range dependencies. However, the vision transformer (ViT)³⁵ has emerged as a solution, overcoming CNN limitations in medical imaging tasks.^{31,36–38} Combining CNN with ViT has improved mammogram analysis, surpassing previous benchmarks.³⁹ Lee et al.⁴⁰ further refined this by incorporating both prior and current mammograms in the CNN-Transformer architecture, enhancing the model's ability to analyze mammogram risk factors by capturing both detailed local and global features.

To overcome existing challenges, we introduce a technique that combines a Transformer and CNN to efficiently integrate global and local feature analysis. Inspired by the diagnostic process of physicians, who compare prior and current mammogram images to detect changes over time, our proposed model successfully incorporates these temporal views to accurately differentiate between normal and malignant breast conditions. The suggested model exhibits reliable classification performance, attaining outcomes that are competitive within the domain. Furthermore, to evaluate the efficacy of the proposed model, we conduct a comprehensive comparison against several baseline models. All selected baselines incorporate both current and prior mammogram images for analysis, reflecting a comprehensive approach for evaluating changes over time. The key contributions are:

1. This paper introduces an end-to-end model that combines CNN and Transformer for breast cancer abnormalities prediction from current and prior mammograms. The model leverages both local and global features to enhance diagnosis ACC.
2. We include a novel position-wise feedforward network (FFN) that incorporates multi-head self-attention (MSA) within a Transformer architecture to learn complex relationships and detect subtle changes between sequential yearly mammograms.
3. The proposed model utilizes positional encoding (PE) to precisely encode the sequential positioning of features from both prior and current mammograms, and channel attention (CA) to highlight key features, sharpening the distinction between normal and cancerous tissues.
4. We employ focal loss (FL) to prioritize instances that are more challenging to classify as either normal or cancer. This technique aims to reduce the rate of false negatives, thereby enhancing our model's ability

to detect subtle signs of breast cancer with increased ACC.

2 | METHODOLOGY

Our research introduces a novel architecture that combines the capabilities of CNNs with the Transformer model, including sophisticated attention mechanisms. By integrating these approaches, the model aims to capture both the local features of individual images and the intricate relationships between paired images (prior and current) that are crucial for identifying subtle changes over time. This is particularly relevant in the context of breast cancer screening, where the ability to detect minute abnormal changes can be the difference between early intervention and a missed diagnosis.

Our approach utilizes pairs of current and prior mammogram images that can be represented as follows:

$$S = \left\{ \left(\mathbf{X}_i^p, \mathbf{X}_i^c, y_i \right) \right\}_{i=1}^N,$$

denoting a paired of current and prior mammogram images dataset of size N , where $\mathbf{X}_i^p$ and $\mathbf{X}_i^c$ represent the i -th prior and current medical images, respectively, and y_i is the corresponding binary category label. Each image $\mathbf{X}_i^p, \mathbf{X}_i^c \in \mathbb{R}^{H \times W \times C}$, with H , W , and C denoting the height, the width, and the number of channels, respectively. For compatibility with our proposed architecture, optimized for three-channel RGB input, we replicate the grayscale data across all three channels, transforming each image into $\mathbb{R}^{H \times W \times 3}$. The label $y_i \in 0, 1$ indicates whether the sample is normal 0 or has cancer 1.

The model employs shared-weight of modified ResNet50⁴¹ models for feature extraction from both prior and current images to ensure that the model learns a consistent representation, crucial for comparing images over time. After feature extraction, the model concatenates the features from both images and applies PE to incorporate spatial information, enhancing the model's focus on relevant features. The concatenated features now enriched with positional information, are processed through two Transformer encoder (TE) blocks. These blocks utilize self-attention mechanisms with position-wise FFN to represent information by enabling interactions among different positions via dense layers, allowing the model to focus on the most relevant information for diagnosis. To further refine the features, a CA block is applied. This block emphasizes features that are more indicative of cancer, enhancing the sensitivity (SEN) of the detection process. Finally, the processed features are fed into a classification layer, which outputs a prediction indicating the presence or absence of cancer based on the analysis of the image pair. The comprehensive design of our suggested model is illustrated in Figure 1. In our dataset and

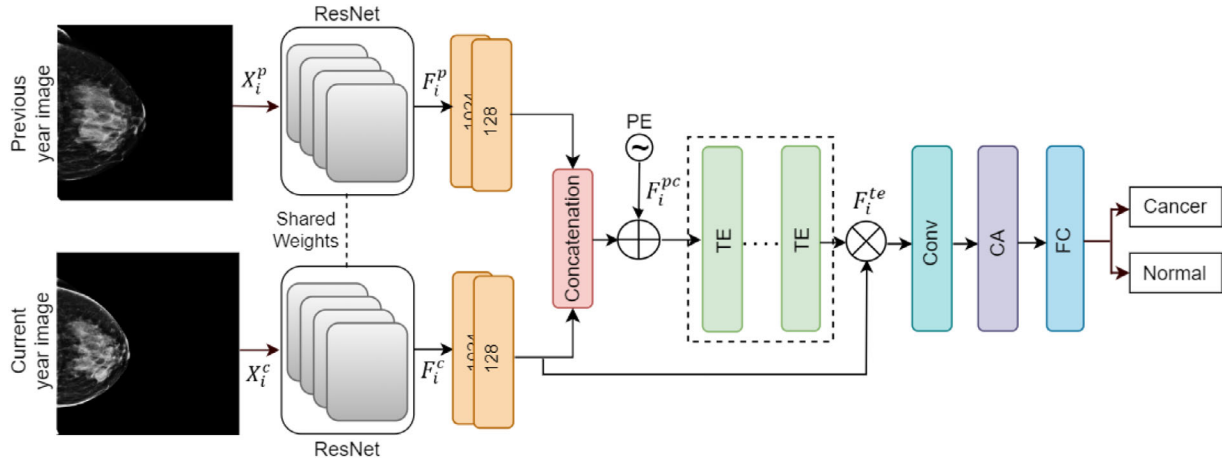


FIGURE 1 Overall architecture of the proposed model. It utilizes a pair of ResNet networks with shared weights to process both the previous year and current year mammogram images simultaneously. Features extracted from both images (denoted as F_i^p and F_i^c) are then concatenated along with PE to capture temporal information. This combined feature set is passed through a TE, further processed by a Conv layer, and then through a CA module to refine feature representation. The output is finally directed through a FC layer for the binary classification task. Here, $\oplus$ and $\otimes$ represent summation and concatenation, respectively. CA, channel attention; Conv, convolutional; FC, fully connected; PE, positional encoding; TE, transformer encoder.

architectural design, our model processes all standard mammographic views (left craniocaudal view (LCC), right craniocaudal view, left mediolateral oblique view, right mediolateral oblique view (RMLO) for both prior and current images, pairing corresponding views (e.g., prior LCC with current LCC).

2.1 | Feature extraction and encoding

The feature extraction and encoding process is a critical component of our architecture, designed to capture and enhance the information embedded within the medical images. This process is articulated through several key equations, as follows:

$$F_i^p = \text{ResNet50}(X_i^p), \quad F_i^c = \text{ResNet50}(X_i^c) \quad (1)$$

Equation (1) describes the initial feature extraction step, where X_i^p and X_i^c represent the prior and current medical images, respectively. We apply the modified ResNet50 model to each image independently to obtain their feature representations: F_i^p for the prior image and F_i^c for the current image. This step is crucial for capturing the high-level features from each image, which are essential for the subsequent analysis.

$$F_i^{pc} = \text{Concatenate}(F_i^p, F_i^c) \quad (2)$$

In Equation (2), the feature representations of the prior and current images are concatenated to form a unified feature vector F_i^{pc} , enabling the model to simultaneously consider both time points. This strategic integration allows the model to directly compare the two sets of features, identifying discrepancies that may

signify temporal changes, such as the development or growth of lesions. The model detects these temporal changes by analyzing variations in the concatenated feature vector, which often represent the progression of disease, like tumor growth or new calcifications (Cals)—key indicators in breast cancer diagnosis.

$$F_i^{pe} = F_i^{pc} + PE \quad (3)$$

Equation (3) introduces PE to the concatenated feature vector F_i^{pc} , resulting in an enhanced feature vector F_i^{pe} . The PE adds critical spatial information to the features, which is essential for the model to interpret spatial relationships and context within the images. By incorporating positional information from both prior and current images, the PE module enhances the representation of spatial features, improving the model's ability to detect subtle changes over time.

$$F_i^{te} = \text{TEBlocks}(F_i^{pe}) \quad (4)$$

Following PE, the feature vector F_i^{pe} is processed through TE blocks (Equation 4), where self-attention mechanisms dynamically weigh and integrate information across various image regions. This approach enables the model to implicitly capture complex relationships and subtle temporal changes between prior and current mammograms.

2.2 | TE block

The TE block is a fundamental element of our architecture, designed to process input sequences through several transformational stages to extract a rich feature

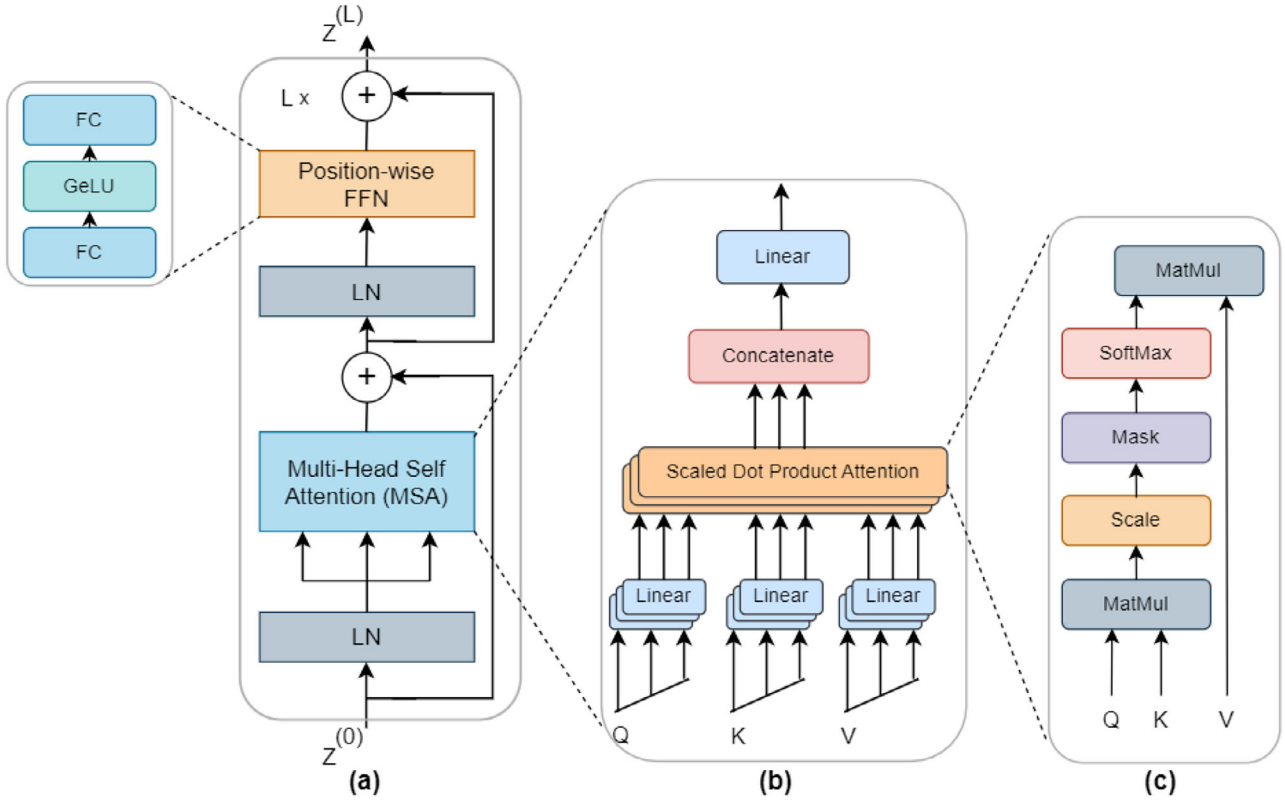


FIGURE 2 The diagram outlines the proposed architecture of a TE block, detailing the MSA mechanism with position-wise FFN and individual self-attention heads. Part (a) shows the overall encoder structure, with layers for MSA, LN, and position-wise FFN, plus RCs. Part (b) expands on the MSA component, illustrating the process of linear transformation of queries (Q), keys (K), and values (V) followed by concatenation. Part (c) zooms into a SA, depicting the sequence of operations: scaling, masking, softmax activation, and matrix multiplication to compute attention scores. FFN, feed-forward networks; LN, layer normalization; MSA, multi-head self-attention; SA, single self-attention head; TE, transformer encoder.

representation. We represent the initial input to the TE block as $Z^{(0)} = F_i^{pe}$, which includes the feature representation F_i^{pc} of the prior and current images, enhanced with PE PE . As shown in Figure 2, each encoder block $l \in \{1, \dots, L\}$ comprises two main subcomponents: multi-head self attention (MSA) and position-wise FFN.⁴² Unlike traditional FFN, our FFN adapts the processing on a position-by-position basis, allowing for more precise modulation of the response based on the positional context within the sequence. Each subcomponent begins with layer normalization (LN) and includes a residual connection (RC), enhancing stability and learning efficiency in deep network structures.

The first subcomponent, MSA, performs the following operation:

$$Z^{(l)'} = \text{MSA}(\text{LN}(Z^{(l-1)})) + Z^{(l-1)} \quad (5)$$

Here, $Z^{(l)'}$ represents the output of the MSA subcomponent for block l . The MSA mechanism enables the model to attend to different positions of the input sequence, capturing the global dependencies within the sequence. The MSA mechanism operates as follows:

For each head i in the range 1 to h , calculate $Q_i = Z^{(l-1)} W_i^Q$, $K_i = Z^{(l-1)} W_i^K$, and $V_i = Z^{(l-1)} W_i^V$. Compute the attention scores as:

$$\text{Attention}(Q_i, K_i, V_i) = \text{softmax}\left(\frac{Q_i K_i^T}{\sqrt{d_k}}\right) V_i \quad (6)$$

Concatenate the outputs of each head and linearly project:

$$\text{MSA}(Z^{(l-1)}) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O \quad (7)$$

where Q_i , K_i , and V_i are the query, key, and value matrices for head i ; $Z^{(l-1)}$ is the input from the previous layer; W_i^Q , W_i^K , and W_i^V are the weight matrices for the queries, keys, and values, respectively; W^O is the weight matrix for the output linear projection; head_i is the output of head i ; and d_k is the dimensionality of the K. The scaled dot-product attention for each head is given by:

$$\text{head}_i = \text{Attention}(Q_i, K_i, V_i) \quad (8)$$

Then the second subcomponent, the position-wise FFN, applies two linear transformations with a Gaussian error linear unit (GeLU) nonlinearity in between:

$$\mathbf{Z}^{(l)} = \text{FFN}(\text{LN}(\mathbf{Z}^{(l)'})) + \mathbf{Z}^{(l)'} \quad (9)$$

where the FFN is described as:

$$\text{FFN}(\mathbf{x}) = \text{FC}_2(\text{GeLU}(\text{FC}_1(\mathbf{x}))) \quad (10)$$

This position-wise FFN enhances the representation capability by enabling interactions among positions through dense layers. Figure 2 provides a thorough representation of the TE module.

After processing through L layers, the output is subjected to a final layer normalization to produce the comprehensive feature representation $\mathbf{F}_i^{(te)}$:

$$\mathbf{F}_i^{(te)} = \text{LN}(\mathbf{Z}^{(L)}) \quad (11)$$

This representation $\mathbf{F}_i^{(te)}$ encapsulates the refined information extracted from the input, integrating both prior and current image features, and is ready for further processing.

2.3 | RC and proposed attention mechanisms

The utilization of a RC is a crucial element when it is concatenated with the output of the TE. The RC ensures that the Transformer does not lose track of these subtle but significant features that might be present in the current year's image. On the other hand, the CA mechanism enhances the model's capacity to identify subtle signs of cancer by highlighting key channels and suppressing less helpful ones. These mechanisms work in the following manner:

2.3.1 | Inclusion of RC

Following the feature extraction and encoding, our architecture incorporates a RC that merges the feature map from the current image with the output of the TE Blocks to enhance the feature representation with fine-grained details from the current input:

$$\mathbf{F}_i^{(merged)} = \text{Concatenate}(\mathbf{F}_i^{(te)}, \mathbf{F}_i^{(c)}) \quad (12)$$

Here, $\mathbf{F}_i^{(merged)}$ represents the combined feature map that integrates both the high-level contextual information from the TE Blocks and the detailed features from the current image, providing a comprehensive feature representation for further processing.

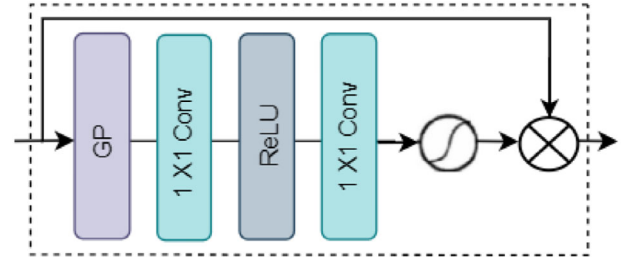


FIGURE 3 The figure illustrates the proposed CA module, which begins with GP and is then followed by two 1×1 Conv layers separated by a ReLU activation function. The module concludes with a sigmoid gate that adjusts the input features by element-wise multiplication. CA, channel attention; Conv, convolutional; GP, global pooling.

After that, $\mathbf{F}_i^{(merged)}$ is subjected to a convolutional layer to further refine the features. This convolutional step is crucial for enhancing local features and preparing the data for the next CA Block.

2.3.2 | CA block

Upon processing through the convolutional layer, the feature map enters the proposed CA Block, which aims to emphasize the most informative channels. The proposed CA block employs 1×1 convolution, utilizing fewer parameters and enabling the model to focus on relevant patterns indicative of cancerous tissues. This is achieved with less computational overhead than would be required by larger convolutional kernels. In Figure 3, the proposed CA block is illustrated. The CA block involves a series of activities in the following order:

At first, a GP layer compresses the spatial dimensions of the feature map, yielding a channel descriptor. The global pooled feature map is then transformed by a sequence of operations, starting with a rectified linear unit (ReLU) activation applied to a 1×1 convolution:

$$\mathbf{F}_i^{(ca)'} = \text{ReLU}(\text{Conv}_{1 \times 1}(\mathbf{F}_i^{(gp)})) \quad (13)$$

$$\mathbf{F}_i^{(ca)''} = \text{Conv}_{1 \times 1}(\mathbf{F}_i^{(ca)'}) \quad (14)$$

The sigmoid function (σ) is applied to each scalar value within the feature map $\mathbf{F}_i^{(ca)''}$, transforming it into a channel-wise attention map (Equation 15). Although each input to the function is a scalar, the output is a matrix of the same size as the input feature map. This matrix, denoted as $\mathbf{F}_i^{(scaled)}$, represents the importance of each channel, with each element scaled between 0 (least important) and 1 (most important), providing a mechanism for focused feature enhancement.

$$\mathbf{F}_i^{(scaled)} = \sigma(\mathbf{F}_i^{(ca)''}) \quad (15)$$

As expressed in Equation (16), the attention map is then element-wise multiplied with the convolutional feature map. In order to improve the feature map for subsequent classification, this stage selectively emphasizes features that the attention mechanism considers relevant while decreasing the remaining features.

$$\mathbf{F}_i^{(ca)} = \mathbf{F}_i^{(scaled)} \odot \mathbf{F}_i^{(conv)} \quad (16)$$

Finally, the channel-refined feature map $\mathbf{F}_i^{(ca)}$ is subsequently inputted into a fully connected (FC) layer, and the final predictions are made by a softmax activation function (Equation 17). This last step classifies the inputs by probabilistically determining the presence of features that indicate cancer:

$$\hat{y}_i = \text{softmax}\left(\text{FC}\left(\mathbf{F}_i^{(ca)}\right)\right) \quad (17)$$

2.4 | Loss function

Our proposed model employs a composite loss function designed to optimize the model's performance while also incorporating regularization to prevent overfitting. The total loss function, $\mathcal{L}$, is formulated as:

$$\mathcal{L} = \alpha \mathcal{L}_{\text{focal}} + \beta \mathcal{L}_{\text{norm1}} + \gamma \mathcal{L}_{\text{norm2}}^2, \quad (18)$$

where each component serves a distinct purpose in the training process.

The FL,⁴³ $\mathcal{L}_{\text{focal}}$, is central to addressing class imbalance by modifying the CEL such that it places more focus on hard-to-classify examples. To complement the FL, we incorporate regularization through the L1 norm ($\mathcal{L}_{\text{norm1}}$) and the squared L2 norm ($\mathcal{L}_{\text{norm2}}^2$)³⁴ of the model parameters. Both techniques can prevent overfitting and enhance the generalization of the model. The coefficients α, β , and γ serve to balance the contributions of FL, L1 norm, and L2 norm squared, respectively, allowing for fine-tuning of the model's SEN to each aspect of the loss function. These hyperparameters are crucial for tailoring the model to specific datasets and objectives, and their optimal values are typically determined through cross-validation.

2.5 | Baseline models

In order to thoroughly evaluate the efficacy of the proposed model, we conducted comprehensive comparison assessments against eight benchmark models, including well-known ones, such as the deep learning-based ResNet50,⁴¹ as well as specialized models specifically designed for the Vanilla Siamese,⁴⁴ which is developed for one-shot image recognition, the prior and current mammogram analysis developed by Bai et al.,³⁴ Lee et al.,⁴⁰ and a multiview analysis²⁷ model developed

by Xia et al.³⁰ for classifying benign and malignant. To ensure a thorough and fair assessment, we customized these models to align with our dataset's characteristics and layer configurations to accurately reflect the processing pipeline of our study. Initially, we used a pre-trained ResNet model, originally developed by He et al.⁴¹, and modified its architecture to serve as the backbone for our proposed model as well as all the baseline models for feature extraction. These modifications included resizing the input layer to accommodate 1024×1024 pixel images and reconfiguring the convolutional layers to process high-resolution mammograms. Additionally, we progressively reduced feature dimensions from $512 \times 512 \times 64$ to $32 \times 32 \times 2048$, and added a specialized transformation layer that converts the $32 \times 32 \times 2048$ feature map to $32 \times 32 \times 128$. These modifications are compatible with the pre-trained weights and were further refined through additional pre-training on a mammogram dataset, ensuring that the model is specifically tailored for mammography analysis.

To ensure consistency in our evaluation, we applied the modified backbone across all models, including both our proposed model and the baseline models. In our comparison, ResNet50 and Sarker et al. were trained only on current images, while other models incorporated both current and prior images. This comparison highlights the significance of using historical information, as the models using only current images showed lower ACC. We used ImageNet weights to initialize the modified backbone, providing a solid foundation for further fine-tuning on our mammogram dataset. However, the baseline models are briefly described below:

1. SMResNet50: Extends ResNet50 by adding two FC layers with dimensions of 512 and 256 for binary classification.
2. Sarker et al.⁴⁵: Employs a shifted window-based dynamic attention block within a transformer architecture to integrate multiview information, enhancing spatial feature coherence across current mammogram images.
3. Vanilla Siamese⁴⁴: Utilizes Euclidean distance and contrastive loss for discrimination.
4. Feature Fusion Siamese Network³⁴: Combines Euclidean and pixel-wise distance metrics, followed by sigmoid activation and binary CEL.
5. PRIME⁴⁰: Merges current and previous mammogram images without using a TD.
6. Yala et al.³⁹: Combines current and previous mammogram images with our modified backbone as the image encoder and utilizes a transformer for aggregating the image features.
7. PRIME+⁴⁰: Adds a more intricate approach, combining features from current and previous mammogram images analyzed with TDs, and incorporating RC.
8. Xia et al.³⁰: Utilizes 12 TEs for multiview classification and a multiplex convolutions module for detailed local

TABLE 1 Data sets information: Overview of the five datasets used in the study, including their respective locations, imaging modalities, resolutions, and the number of normal and cancer cases.

Data set	Location	Modality	Resolution	History cases	Normal cases	Cancer cases
DDSM ⁴⁶	USA	2D	3000 × 4800	×	—	2,055
CMMD ⁴⁷	China	2D	1914 × 2294	×	—	2632
BCS-DBT ⁴⁸	USA	3D	1890 × 2457	×	8528	75
VinDr-Mammo ⁴⁹	Vietnam	2D	3518 × 2800	×	—	1,825
UCHC	USA	2D	5928 × 4728 to 2294 × 1914	✓	815 (pairs)	627 (pairs)

Abbreviations: BCS-DBT, breast cancer screening-digital breast tomosynthesis; CMMD, Chinese mammography database; DDSM, digital database for screening mammography; UCHC, University of Connecticut Health Center.

feature extraction. Their model used the CC and MLO views, while we employed the CC and MLO views of both previous and current images.

3 | MATERIALS

3.1 | Datasets

In our study, we utilized a composite of five separate datasets, with one dataset exclusively used to train and test the proposed and baseline models for our comprehensive analysis, and the remaining four allocated for pretraining the backbone model. The primary dataset, obtained from the Radiology Department of the University of Connecticut Health Center (UCHC), consists of both current and historical mammograms and was used to validate and enhance our model, as well as to assess the efficacy of the baseline models. For the backbone model, images from the following four datasets were combined and standardized: (1) the Digital Database for Screening Mammography (DDSM),⁴⁶ (2) the Chinese Mammography Database (CMMD),⁴⁷ (3) the Breast Cancer Screening-Digital Breast Tomosynthesis (BCS-DBT),⁴⁸ and (4) VinDr-Mammo (VDM).⁴⁹ For a summary of the datasets, see Table 1.

3.1.1 | UCHC dataset

The study employed a dataset from the UCHC, comprising both current and historical mammograms for training, validating, and testing the models. The dataset includes FFDM from 381 patients aged 28 to 95, categorized as follows: 146 with masses, 84 with AD, 90 with MCs, and 61 normal. The images were captured with Hologic mammography systems and annotated by radiologists.

The dataset includes 627 pairs labeled as cancerous, with current cancerous FFDMs and their corresponding prior normal FFDMs, and 815 pairs labeled as normal, where both current and prior FFDMs are normal. Approximately 83.4% of patients had follow-up visits within 1 to 3 years. Cancer cases were confirmed via biopsy and are identifiable in both Cranio-Caudal (CC)

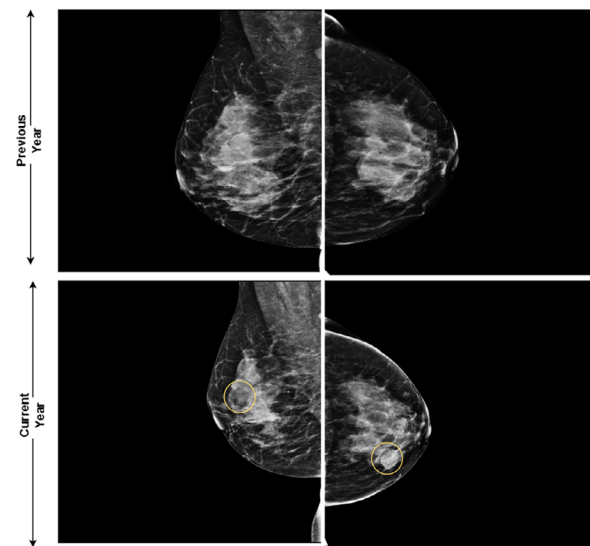


FIGURE 4 UCHC Dataset Visualization showcasing pairs of mammogram images. The top row displays previous year mammograms, while the bottom row presents current year mammograms with cancerous tumors highlighted by yellow circles. UCHC, University of Connecticut Health Center.

and MLO views, receiving The breast imaging reporting and data system (BI-RADS) scores of 4 or 5. Normal cases exhibited no abnormalities, with BI-RADS scores of 1 or 2. A sample (a cancer pair of current and previous year images) of UCHC dataset is depicted in Figure 4.

3.1.2 | Public datasets for training the backbone

The DDSM dataset⁴⁶ is a well-known resource for mammography research, including normal, benign, and malignant cases. For this study, we used 2055 cancer cases, with original mammograms averaging 3000 × 4800 pixels. The CMMD dataset⁴⁷ contains both benign and malignant cases. We focused on 2632 cancer cases, with original mammograms averaging 1914 × 2294 pixels. The BCS-DBT dataset⁴⁸ provides normal and cancerous cases in 3D. We synthesized 2D mammograms (s2D) from 3D images, obtaining 8528

normal s2D and 75 cancer s2D cases for this study. VDM⁴⁹ comprises 20 000 scans from 5000 FFDM studies. For this study, we used 1825 images rated as BI-RADS 4 to 5, averaging 3518×2800 pixels. To align with the focus of our study, we excluded benign cases from all datasets.

3.2 | Dataset preprocessing

To prepare the training dataset for pretraining the backbone model, images from four distinct sources: DDSM, CMMD, 2D BCS-DBT, and VDM mammograms (see Table 1) are combined. The dataset is standardized by normalizing all images according to the formula outlined by ref³⁴.

The UCHC dataset preprocessing included cropping using the breast region segmentation (BRS) module⁵⁰ to delineate and isolate breast tissue. The BRS module operates by first detecting the breast region within the mammogram image. Once detected, the module crops this selected region to exclude any background or irrelevant areas. The cropped image is then resized to a square shape by adding padding, ensuring that the original breast region remains intact and uniform across all images fed into the model. This preprocessing workflow ensures that the model focuses solely on the breast tissue, which is essential for accurate and efficient analysis in the subsequent stages of processing and modeling. Following BRS, the images were resized to a uniform dimension of 1024×1024 pixels. Additional preprocessing steps included morphological operations for removing annotations, the application of masking to improve clarity, and contrast limited adaptive histogram equalization (CLAHE) to enhance visibility in dense tissue areas. Median filtering and unsharp masking techniques were also applied to decrease noise and enhance important diagnostic features like MCs and masses, respectively. This comprehensive preprocessing approach improved the quality of the images, making them more suitable for precise analysis and reducing the computational load. However, to evaluate the effectiveness of these preprocessing techniques, we conducted a quantitative analysis, presented in Table S1, which shows the impact of different preprocessing combinations on model ACC. Additionally, Figure S1 provides visualizations of the preprocessing effects on mammogram images before and after applying each technique. These results underscore the importance of our preprocessing steps in enhancing image quality and improving model ACC.

3.3 | Data augmentation

To enhance the diversity and robustness of our training dataset, we employed a series of data augmentation

techniques aimed at introducing variability and complexity. Specifically, we used random rotations, allowing images to be rotated by either 90 or 180 degrees, to introduce greater variability into the dataset. This strategy was designed to enhance the model's ability to generalize across a range of imaging scenarios, thereby improving its robustness in accurately categorizing mammograms. Additionally, we applied random horizontal and vertical flips to simulate mirror images, increasing the dataset's diversity. To account for positional variations, we introduced random translations, shifting images up to 10% in any direction. Color adjustments through slight modifications to brightness, contrast, saturation, and hue were also applied to mimic different lighting conditions and color variations. Finally, normalization of the images was performed to ensure a consistent distribution of pixel values, facilitating more efficient model training.

4 | EXPERIMENTAL SETUP

4.1 | Training configuration

To mitigate the risk of overfitting, our approach incorporated transfer learning techniques for both the proposed and baseline models' training processes. Utilizing the modified ResNet50 architecture for the model backbone network, which had been previously trained on the ImageNet dataset.⁵¹ Subsequently, this model was further pretrained on a composite mammogram dataset (see data preprocessing for details) to refine the models specifically for mammography image analysis. While transfer learning leverages pre-trained features, it does not fully prevent overfitting. To mitigate this, we froze early convolutional layers during fine-tuning. This approach reduces overfitting, lowers computational costs, and enhances performance by utilizing pre-learned features. The detailed workflow of this training procedures is illustrated in Figure 5. The refined networks then served as the backbone for both our proposed models and the baseline models,^{30,34,40,44} which are subsequently trained on the UCHC dataset. For further details, please refer to the "Baseline Models" subsection. To ensure consistent evaluation, we used five-fold cross-validation to evaluate model performance, consistently applying this method across all baseline models. The dataset was split into five subsets, with the model trained on four and validated on one, repeated five times. Data augmentation was applied only to the training set to enhance diversity and durability.

Throughout the 60 epochs, we conducted a comprehensive examination of various hyperparameters. We investigated starting learning rates within the spectrum of 1×10^{-2} to 1×10^{-5} , employing a cosine learning rate scheduler to refine the learning rate over time. Given the computational constraints associated with

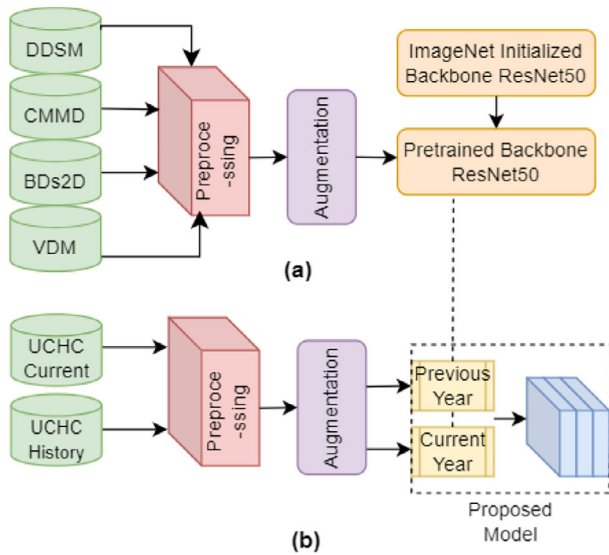


FIGURE 5 Training workflow for the proposed model. (a) The model begins with a ResNet50 backbone that has been pretrained on the ImageNet dataset. This backbone is then further pretrained on the DDSM, CMMD, BDs2D, and VDM datasets. The pretrained weights are transmitted to the backbone of our proposed model, using the shared weights of ResNet. (b) The pretrained ResNet is further optimized by fine-tuning it using paired mammograms from both the current and earlier years of the UCHC dataset, as part of our proposed model. BCS-DBT, breast cancer screening-digital breast tomosynthesis; BDs2D, BCS-DBT; CMMD, Chinese mammography database; DDSM, digital database for screening mammography; UCHC, University of Connecticut Health Center.

processing high-resolution images, we adopted a strategy of end-to-end mini-batch stochastic gradient descent, with a batch size of 4. To this end, we incorporated dropout rates ranging from 0.2 to 0.5 in the FC layers. Additionally, we applied L1 norm (with a β value ranging from 1×10^{-5} to 2) and L2 norm regularizers (with a γ value ranging from 1×10^{-5} to 2) within the FC layers for further regularization. To assess the effectiveness of FL, the α parameter was varied within a scope ranging from 0.25 to 0.75. The models' ACC was optimized using the ADAM optimizer, renowned for its adaptive momentum capabilities. All models were trained and evaluated on Tesla V100 GPUs, each equipped with 32 GB of memory to accommodate the demands of high-resolution input data.

4.2 | Evaluation metrics

In our study, we utilized the UCHC dataset for the purpose of assessing the effectiveness of our proposed model and also the baseline models. In evaluating the models, we focus on six critical metrics: ACC, recall (or sensitivity [SEN]), precision (PRE), specificity (SPE), F1-score, and AUC (Area Under the ROC Curve).^{30,34}

Moreover, to provide a comprehensive understanding of the model's performance, we also consider the false

positive rate (FPR), false negative rate (FNR), false discovery rate (FDR), and emphasize the importance of the true positive rate (TPR).³⁴

5 | RESULTS

5.1 | Performance analysis

The comprehensive evaluation presented in Table 2 offers a detailed comparative analysis of performance metrics for the baseline models, benchmarked against our proposed model on the UCHC private dataset. This analysis encompasses a range of metrics, including ACC, SPE, SEN, PRE, F1 score, and AUC, with $\pm$ values indicating the margin of error or confidence interval. The first two rows depict models that were evaluated only based on current images, while the subsequent models were evaluated using both prior and current images.

The baseline model, SM (ResNet50), presents an ACC of 84.06% and an AUC of 85.87%, illustrating the potential limitations when using a single-image approach in complex diagnostic tasks. The absence of historical images results in a significant loss of diagnostic detail, which is crucial for accurate mammogram interpretation. We also considered another state-of-the-art model for analyzing current images, as proposed by Sarker et al.⁴⁵ Despite leveraging a strong architecture based on transformers, their model achieved an AUC of 86.11% and an ACC of 84.38%. However, since this model used only current images for classifying normal and cancer cases, its performance was limited.

The Vanilla Siamese network, with an ACC of 84.97% and a SEN of 84.20%, highlights the need for more sophisticated methods for analysis, as relying solely on Euclidean distance may overlook complex and subtle patterns in mammograms that are essential for early cancer detection. Lee et al.⁴⁰ provide two models, Prime and Prime+, where the former achieves an ACC of 85.24% and an AUC of 87.91%. Despite the enhanced performance of Prime+ with an ACC of 86.01% and an AUC of 89.94%, it still lacks PE in its design, which raises concerns about the model's ability to fully capture and exploit spatial relationships within mammographic images. Additionally, Prime+ integrates a RC from the current images to the FC layer, but the absence of PE and CA mechanisms in this architecture limits its performance. Yala et al.'s MIRAI architecture,³⁹ which achieved an AUC of 88.09% and an ACC of 85.42%, also lacks PE, CA, and RC, potentially explaining its relatively lower performance compared to our proposed model. While MIRAI directs the output from the image aggregator to a FC layer, our model benefits from the added impact of PE, CA, and RC before this step, leading to improved performance metrics.

The model developed by Bai et al.³⁴ achieves an ACC of 85.67% and an AUC of 89.61%, demonstrating

TABLE 2 A comparative analysis of model performance metrics, showcasing the ACC, SEN, PRE, SPE, F1-score, and AUC for the baseline models including SM (ResNet50), Vanilla Siamese, and models by Lee et al. (Prime and Prime+), Bai et al., Xia et al., Sarker et al., Yala et al., and the proposed model.

Model	ACC	SEN	PRE	SPE	F1-score	AUC
Using only current images						
SM (ResNet50)	84.06% $\pm$ 0.2%	83.60% $\pm$ 0.24%	84.06% $\pm$ 0.2%	84.80% $\pm$ 0.23%	84.00% $\pm$ 0.21%	85.87% $\pm$ 0.2%
Sarker et al.	84.38% $\pm$ 0.11%	83.67% $\pm$ 0.19%	84.09% $\pm$ 0.23%	85.28% $\pm$ 0.11%	84.21% $\pm$ 0.13%	86.11% $\pm$ 0.23%
Using both prior and current images						
Vanilla Siamese	84.97% $\pm$ 0.23%	84.20% $\pm$ 0.23%	84.40% $\pm$ 0.24%	85.30% $\pm$ 0.23%	84.20% $\pm$ 0.21%	87.97% $\pm$ 0.25%
Lee et al. (Prime)	85.24% $\pm$ 0.19%	84.30% $\pm$ 0.18%	85.20% $\pm$ 0.19%	85.11% $\pm$ 0.17%	84.30% $\pm$ 0.19%	87.91% $\pm$ 0.18%
Yala et al.	85.42% $\pm$ 0.13%	84.59% $\pm$ 0.33%	83.74% $\pm$ 0.33%	86.50% $\pm$ 0.19%	84.83% $\pm$ 0.29%	88.09% $\pm$ 0.33%
Bai et al.	85.67% $\pm$ 0.1%	86.74% $\pm$ 0.13%	85.67% $\pm$ 0.1%	85.71% $\pm$ 0.12%	86.74% $\pm$ 0.1%	89.61% $\pm$ 0.11%
Lee et al. (Prime+)	86.01% $\pm$ 0.11%	85.61% $\pm$ 0.11%	86.01% $\pm$ 0.12%	86.50% $\pm$ 0.11%	87.05% $\pm$ 0.13%	89.94% $\pm$ 0.11%
Xia et al.	87.94% $\pm$ 0.2%	86.77% $\pm$ 0.21%	87.60% $\pm$ 0.2%	87.73% $\pm$ 0.21%	87.89% $\pm$ 0.22%	90.26% $\pm$ 0.22%
Proposed model	90.80% $\pm$ 0.011%	90.80% $\pm$ 0.011%	90.80% $\pm$ 0.012%	90.88% $\pm$ 0.011%	90.95% $\pm$ 0.011%	92.58% $\pm$ 0.012%

Note: The highest performance for each metric is in bold.

Abbreviations: ACC, accuracy; AUC, area under the curve; PRE, precision; SM, single model; SEN, sensitivity; SPE, specificity.

improvement by using a Siamese network with distance functions. However, the reliance on cross-entropy loss (CEL) without accounting for the unique characteristics that the Siamese architecture can detect might limit its ability to differentiate subtle diagnostic features. Unlike Bai et al.'s approach, which used simple distance functions, our model employs a TE that leverages self-attention mechanisms. This allows our model to better capture complex relationships and subtle changes between prior and current mammograms, which are often challenging for simple distance metrics, especially when the features are highly correlated. Xia et al.'s³⁰ model achieves an ACC of 87.94% and an AUC of 90.26%. It combines global and local variables, reflecting the benefits of integrating different architectural paradigms. For our comparison, we used only their architectural structure. However, our model uniquely pairs views from different time points, such as prior LCC with current LCC and prior RMLO with current RMLO, thus considering temporal changes similar to Xia et al.'s approach but with a more sophisticated mechanism for capturing temporal and spatial relationships.

Finally, our proposed model exhibits superior performance, with an ACC of 90.80% and an AUC of 92.58%. It distinguishes itself by leveraging not only modified backbone for feature extraction but also integrating PE, which anchors the spatial relevance of image features, enhancing the model's ACC. This approach, alongside the use of two TEs and CA mechanisms, ensures a comprehensive evaluation of mammographic images, culminating in heightened SEN and PRE, both reaching 90.80%. Overall, our results emphasize the capability of our suggested method to greatly improve the PRE of mammography interpretation, opening up possibilities for advancements in the early identification of breast cancer.

Figure 6 compares the balanced accuracy, Area Under the Precision-Recall Curve (AUC-PR), and Brier Score⁵² of baseline models and our proposed model on the imbalanced UCHC dataset. Despite the dataset imbalance, the proposed model demonstrates superior performance in both balanced accuracy and AUC-PR, along with a lower Brier Score, further underscoring its effectiveness in mammogram classification. For more details, please refer to the supplementary material (Table S2), and the confusion matrix for all models is presented in Figure S2.

Figure 7 shows the comparative efficacy of our proposed model against baseline models using ROC curves, validated by five-fold cross-validation. The SM (ResNet50) model achieved an AUC of 0.858 ($p = 0.013$), while Sarker et al.⁴⁵ reached 0.861 ($p = 0.012$), both of which only used current images. In contrast, the models using both prior and current images showed clear performance improvements. For example, the Vanilla Siamese model⁴⁴ and Lee et al. (Prime)⁴⁰ both achieved 0.879 with $p = 0.009$ and $p = 0.007$ values, respectively. Yala et al.³⁹ attained 0.880 ($p = 0.007$) with MIRAI, but it lacked RC, PE, and CA. Prime+ reached 0.899 ($p = 0.006$), incorporating RC. Bai et al.³⁴ scored 0.896 ($p = 0.007$), while Xia et al.³⁰ achieved 0.903 ($p = 0.006$). Our model, leveraging prior and current images with ResNet50, PE, and advanced TE with CA, improved to 0.926, highlighting the value of historical data for better classification.

Table 3 presents a comprehensive comparison of the FDR and FNR across the baseline models with our proposed model used in mammography analysis for various kinds of abnormalities, including Mass, Cals, and AD. Lower FDR and FNR values indicate fewer false-positive and false-negative errors, respectively, in identifying these types of abnormalities. Our proposed

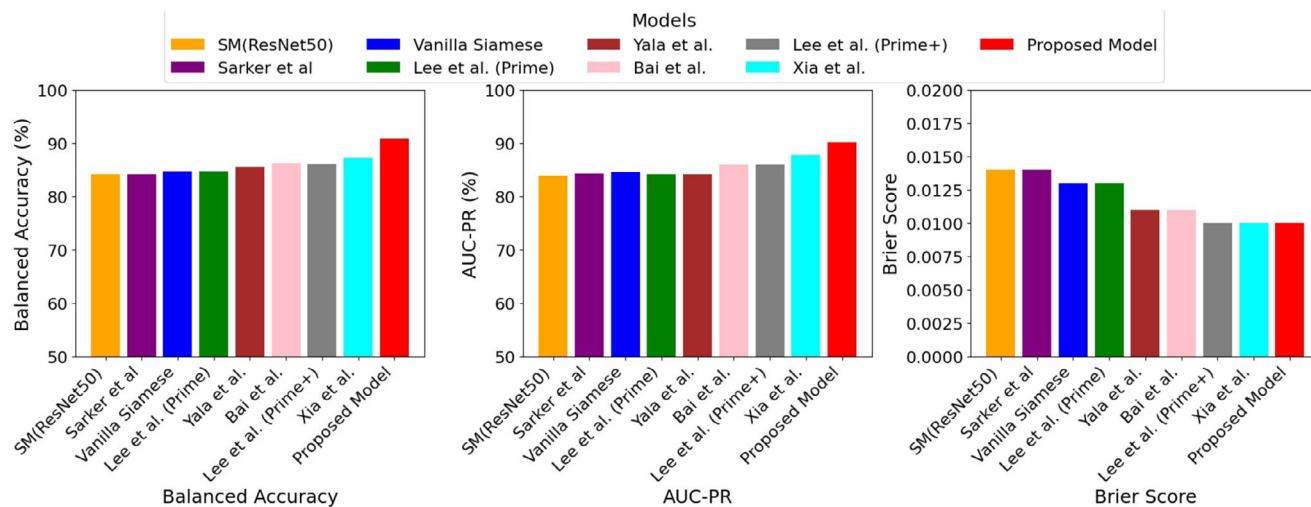


FIGURE 6 Comparison of model performance across balanced accuracy, AUC-PR, and Brier Score metrics.

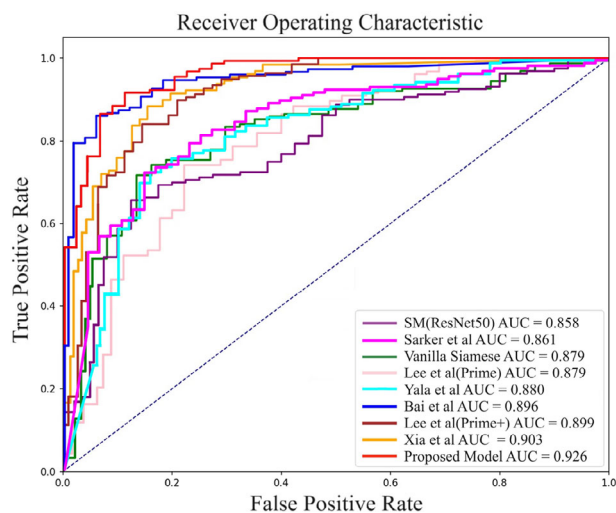


FIGURE 7 The graph represents the ROC curves for various baseline models, comparing their TPR against FPR. FPR, false positive rates; ROC, receiver operating characteristic; TPR, true positive rates.

model has the lowest error rates for masses, with a FDR of 0.14 and a FNR of 0.09. This implies superior performance in accurately diagnosing mass anomalies compared to previous models. The SM (ResNet50) has the highest FDR and FNR in mass detection, indicating a greater probability of misclassifications. This highlights the need to include prior image comparisons to improve ACC.

For the more challenging task of detecting Cals and AD, the proposed model outperforms competing ones with the lowest rates of false discoveries (FDR of 0.20 for Cals and 0.48 for AD) and false negatives (FNR of 0.08 for Cals and 0.22 for AD). This illustrates the model's reliability in accurately identifying Cals and AD while minimizing misdiagnosis. Therefore, the accurate identification accomplished by this system is attributed

TABLE 3 Comparison of FDR and FNR metrics with the baseline models, including SM (ResNet50), Vanilla Siamese, and others by Lee et al., Bai et al., Xia et al., as well as the proposed model, across three categories: Mass, Cals, and AD in mammogram analysis.

Model	Mass		Cals		AD	
	FDR	FNR	FDR	FNR	FDR	FNR
Using only current images						
SM (ResNet50)	0.28	0.20	0.40	0.19	0.80	0.53
Sarker et al.	0.28	0.19	0.33	0.14	0.72	0.47
Using both prior and current images						
Vanilla Siamese	0.25	0.18	0.32	0.15	0.73	0.47
Lee et al. (Prime)	0.23	0.17	0.30	0.14	0.69	0.45
Yala et al.	0.16	0.15	0.28	0.11	0.51	0.38
Bai et al.	0.17	0.11	0.26	0.09	0.55	0.30
Lee et al. (Prime+)	0.17	0.11	0.22	0.09	0.53	0.25
Xia et al.	0.16	0.10	0.22	0.08	0.51	0.25
Proposed model	0.14	0.09	0.20	0.08	0.48	0.22

Note: The highest performance for each metric is in bold.

Abbreviations: AD, architectural distortion; Cals, calcifications; FDR, false discovery rate; FNR, false negative rate.

to the employment of sophisticated components in its design. Specifically, a modified backbone is employed for precise feature extraction, PE is used to incorporate spatial context, and a pair of TEs, enhanced by CA mechanisms, are employed to finely adjust the model's focus on relevant features. This combination guarantees a thorough and subtle investigation, leading to its exceptional performance.

To assess the interpretability of our proposed model, we applied gradient-weighted class activation map (Grad-CAM)⁵³ to generate heatmaps that highlight cancerous regions. It is important to note that Grad-CAM heatmaps highlight the activated discriminatory parts of the images that strongly influence the model's decision-making process. In these heatmaps, blue and green

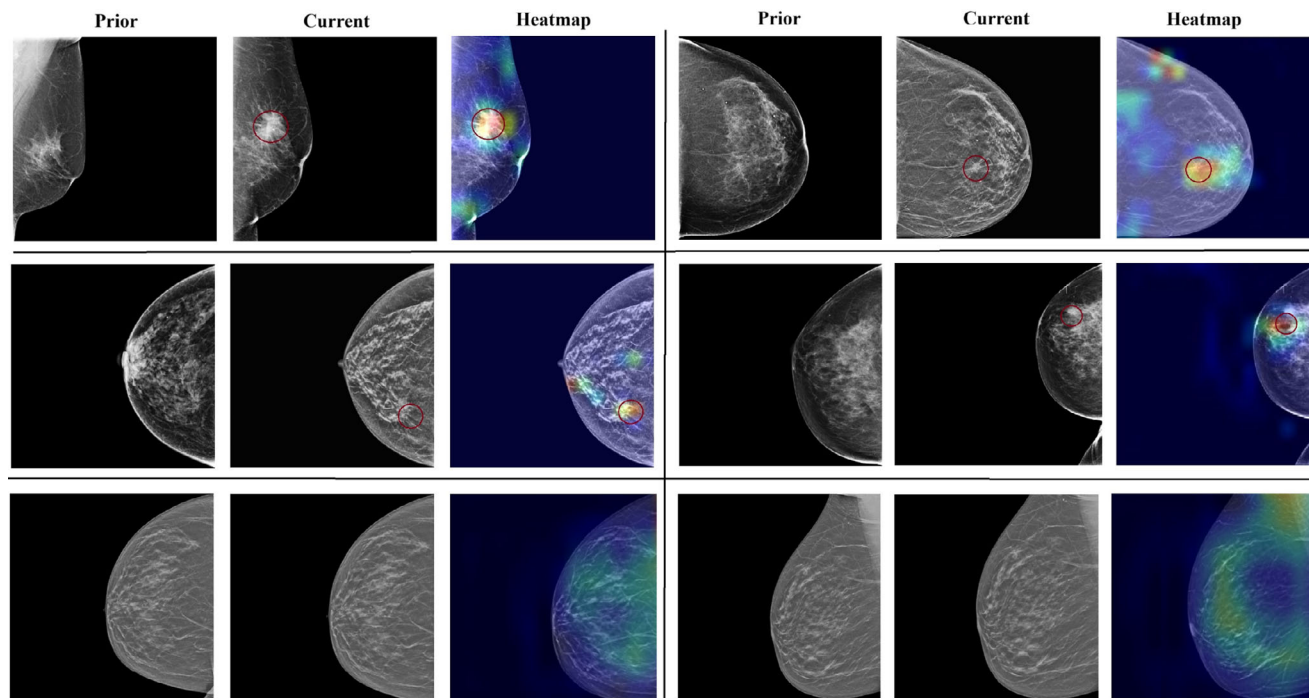


FIGURE 8 Visualization of Grad-CAM heatmaps applied to prior and current mammogram images. The first two rows depict four cancer cases from four different patients, with red circles indicating specific cancerous areas, while the last row represents two normal cases from two other patients. Grad-CAMs, gradient-weighted class activation maps.

areas represent less relevant regions, while red indicates the most critical areas that strongly impact the model's predictions. These visualizations provide valuable insights into how the model leverages spatial and temporal changes across mammograms to enhance cancer detection. Figure 8 displays prior images, current images, and the corresponding Grad-CAM heatmaps of six patients, with the first two rows presenting cancer cases from four different patients and the third row showing normal cases from two other patients. The red circles indicate specific cancerous areas as annotated by radiologists. As can be seen in Figure 8, normal cases show no red zones, indicating the absence of cancerous regions. In contrast, cancer cases exhibit distinct red zones, signifying areas likely associated with cancerous regions.

5.2 | Ablation study

Table 4 compares model ACC across different numbers of TE layers. With a consistent structure including PE, RC, CA, and FC layers, the model with two TE layers achieves the highest ACC at 90.80%. One TE layer yields an ACC of 87.91%, while three layers maintain 90.80%. Adding four layers leads to a slight decline to 90.21%, and five layers further reduce ACC to 89.41%. Thus, two TE layers offer an optimal balance of depth and complexity, effectively capturing critical features without overfitting.

TABLE 4 The performance impact of varying the number of TE layers within a ResNet50-based architecture, also highlighting the presence of PE, RC, CA, and FC layers on the overall ACC.

ResNet50	PE	TE layers	RC	CA	FC	ACC
✓	✓	1	✓	✓	✓	87.91%
✓	✓	2	✓	✓	✓	90.80%
✓	✓	3	✓	✓	✓	90.80%
✓	✓	4	✓	✓	✓	90.21%
✓	✓	5	✓	✓	✓	89.41%

Note: The highest performance for each metric is in bold.

Abbreviations: ACC, accuracy; CA, channel attention; FC, fully connected; PE, positional encoding; RC, residual connection; TE, Transformer encoder.

Table 5 showcases the performance of models using CEL with a ResNet50 backbone, PE, TE, RC, CA, and FC layers. While the best ACC with CEL is 89.98% using two TE layers, our proposed model with FL achieves a higher ACC of 90.80%. This improvement demonstrates the FL's effectiveness in refining the model's focus on hard-to-classify examples, ensuring our model learns efficiently and generalizes better with complex data.

Table 6 highlights the effect of FC layers in our proposed model. With two FC layers, the model achieves optimal performance, boasting an ACC of 90.80%, SEN of 90.80%, PRE of 90.80%, SPE of 90.88%, an F1-score of 90.95%, and an AUC of 92.58%. Increasing the number of FC layers to three or four reduces the performance of these measures, suggesting that while additional layers add complexity, they hinder

TABLE 5 The table displays the standard CEL performance with various configurations of TE layers, including PE, RC, CA, and FC layers.

ResNet50	PE	TE layers	RC	CA	FC	Standard (CEL)	ACC
✓	✓	1	✓	✓	✓	✓	85.22%
✓	✓	2	✓	✓	✓	✓	89.98%
✓	✓	3	✓	✓	✓	✓	89.98%
✓	✓	4	✓	✓	✓	✓	89.06%
✓	✓	5	✓	✓	✓	✓	87.61%

Abbreviations: ACC, accuracy; CA, channel attention; CEL, cross entropy loss; FC, fully connected; PE, positional encoding; RC, residual connection; TE, Transformer encoder.

TABLE 6 The table displays how varying the number of FC layers in the proposed network in terms of ACC, SEN, PRE, SPE, F1-score, and the AUC curve.

FC layers	ACC	SEN	PRE	SPE	F1-score	AUC
2	90.80%	90.80%	90.80%	90.88%	90.95%	92.58%
3	89.92%	90.02%	89.92%	90.07%	90.07%	91.77%
4	88.41%	88.77%	88.41%	88.77%	88.90%	90.59%

Note: The highest performance for each metric is in bold.

Abbreviations: ACC, accuracy; AUC, area under the curve; FC, fully connected; PRE, precision; SEN, sensitivity; SPE, specificity.

TABLE 7 The table summarizes an ablation study, showing the ACC impact of including or excluding key components, such as ResNet50, PE, TE, RC, CA, CEL, and FL within the proposed model.

ResNet50	PE	TE	RC	CA	CEL	FL	ACC
✓	X	X	X	X	✓	X	64.16%
✓	X	✓	X	X	✓	X	86.45%
✓	X	✓	X	✓	✓	X	86.72%
✓	X	✓	✓	X	✓	X	87.07%
✓	X	✓	✓	✓	✓	X	87.89%
✓	✓	✓	X	X	✓	X	88.51%
✓	✓	✓	X	✓	✓	X	89.05%
✓	✓	✓	✓	✓	✓	X	89.98%
✓	✓	✓	✓	✓	X	✓	90.80%

Note: The highest performance for each metric is in bold.

Abbreviations: ACC, accuracy; CA, channel attention; CEL, cross entropy loss; FL, focal loss; PE, positional encoding; RC, residual connection; TE, Transformer encoder.

generalization. The two-layer configuration strikes a balance, effectively combining CNN feature extraction with Transformer sequence modeling to identify patterns indicative of breast cancer in mammograms.

Table 7 presents a comprehensive ablation analysis that analyzes the impact of each component in our proposed model on its overall efficacy, as evaluated by ACC. The components under consideration include the ResNet50 base, PE, TE, RC, CA, CEL, and FL. The table shows that when the proposed model uses

only ResNet50, without any additional components, it achieves an ACC of 64.16%. Sequential addition of components—first TE, then CA, followed by RC, and finally combining all components except PE and FL—leads to ACC of 86.45%, 86.72%, 87.07%, and 87.89%, respectively. These components are instrumental in enabling the model to understand the data better and to ensure that valuable information is carried forward through the network layers. This progression highlights the critical roles of PE and FL in enhancing model performance. Notably, integrating PE with TE significantly boosts ACC to 88.51%. With all components except FL, the model achieves an impressive ACC of 89.98%. These improvements are achieved by concentrating on spatial contexts and relevant feature channels. The peak of the model's ACC is achieved when FL is integrated alongside all other components, reaching a notable ACC rate of 90.80%. This underscores the substantial effect of FL, which is particularly valuable because it guides the model to direct more attention to instances that are harder to classify, ensuring that the learning process is not dominated by easy-to-classify examples.

6 | CONCLUSION

This paper presents a CNN-Transformer based model for breast cancer classification using both prior and current mammograms. This model mimics radiologists by carefully examining changes in mammograms over time, which greatly improves its ability to accurately diagnose cancer. A key advancement of a position-wise FFN and MSA enables the recognition of minor changes in mammograms across consecutive years. The proposed model also incorporates PE and CA methods to effectively emphasize the spatial features, allowing for exact discrimination between normal and cancerous tissues. Additionally, the FL focuses on the more difficult cases, reducing the chance of missing cancer cases and making diagnoses more accurate. The model exhibits outstanding performance, achieving an ACC of 90.80% and an AUC of 92.58%, verified through detailed five-fold cross-validation. Furthermore, it achieves the lowest FDR and FNR across different abnormalities, including masses, calcifications, and ADs. This low error rate demonstrates the model's reliability in accurately diagnosing these abnormalities.

Although the classification of mammograms into three classes—normal, benign, and cancer—as presented by studies, such as^{20,30,54} offers a more nuanced understanding, many studies, including our own,^{34,45,55} concentrate on a binary classification of cancer versus non-cancer. This is because there is an imbalance or a lack of benign cases in the datasets that are currently available. Including benign cases is essential to further enhance the model's ability to distinguish between non-cancerous and cancerous findings, thereby improving

diagnostic ACC. To address this limitation, future work will focus on extending our model to support three classes—normal, benign, and cancer—once benign cases with paired prior and current images become available. This enhancement would enable the model to capture a broader spectrum of clinical insights, offering a more comprehensive approach to breast cancer detection. Additionally, we plan to explore methods for integrating information across different mammographic views, aiming to improve diagnostic ACC at the patient level for even more reliable predictions. Besides, incorporating BI-RADS scores alongside these methods could provide critical insights into risk levels and standardize assessments, aligning model predictions with clinical evaluations.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The public DDSM dataset can be accessed at <http://www.eng.usf.edu/cvprg/mammography/database.html>. The public CMMD dataset is available at <https://wiki.cancerimagingarchive.net/pages/viewpage.action?pagelD=70230508>. The BCS-DBT dataset is also publicly accessible at <https://wiki.cancerimagingarchive.net/pages/viewpage.action?pagelD=64685580> and VindrMammo is accessible at <https://vindr.ai/datasets/mammo>. This study's private data set is accessible to the corresponding author upon a reasonable request.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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